Data Analytics Capstone

Retail Sentiment as a Volatility Early-Warning Signal:

A Machine-Learning Study of StockTwits and U.S. Equity Risk

Interim Report

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QM640: Data Analytics Capstone

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[Current Term] Term

[Date]

GitHub repository (data and code):

https://github.com/JamesSavages/Retail_Sentiment_and_Market_Volatility

Introduction

Background and Context

Financial markets are driven not only by fundamentals but by the collective mood of the people trading them, which is now broadcast in real time on social platforms and financial news. A substantial literature establishes that this visible sentiment carries information about market behaviour. Message-board and microblog activity has been linked to trading volume and volatility (Antweiler & Frank, 2004; Sprenger et al., 2014), and social-media mood has been related to index movements (Bollen et al., 2011). Importantly, the strongest and most consistent evidence is for volatility and trading volume rather than for returns, which are closer to efficient (Audrino et al., 2020).

For the businesses that serve retail investors, including brokerages, robo-advisors, and risk-management desks, the operationally valuable question is not what the crowd feels today but whether today's mood signals which stocks will be volatile tomorrow. Volatility surprises are where retail users are most exposed, and where hedging and liquidity-provision costs are incurred. An accurate early-warning signal for next-day volatility therefore has direct commercial value, informing user risk alerts, position sizing, and liquidity decisions.

Measuring sentiment accurately is itself a research problem. Traditional dictionary-based methods have systematically misclassified financial language (Loughran & McDonald, 2011), whereas transformer-based models such as FinBERT read text in context and have improved financial sentiment classification (Araci, 2019). More recently, large language models have shown an emerging capacity for financial reasoning. For example, Lopez-Lira and Tang (2023) found that GPT-4 can predict market reactions to news headlines without task specific training, although returns appear to decay as adoption spreads, further supporting a focus on volatility rather than returns. A further constraint of methodologies used in this field of study is endogeneity. Given that news causes sentiment and price to change at the same time, it is impossible to tell which one is driving the other. To avoid this, this study adopts a predictive design relating sentiment at day t to volatility on day $t + 1$, consistent with prior work (Antweiler & Frank, 2004; Audrino et al., 2020).

Problem Statement

The objective of this study is to predict next-day realized volatility (Y) for eight high-liquidity United States technology equities, with the ticker-day as the unit of analysis, using retail investor sentiment features (X = news sentiment score, message and article volume as an attention proxy) together with lagged market controls, over the period January 2023 to December 2024. Success is evaluated by out-of-sample forecast error (root mean squared error, RMSE, and the quasi-likelihood loss, QLIKE) and by the risk-adjusted return (Sharpe ratio) of a sentiment-informed trading strategy, net of transaction costs.

Purpose of the Study

The study is primarily predictive, forecasting a continuous risk measure, with an explanatory component that quantifies sentiment's marginal information beyond price history, a classification component that predicts next day return direction, and an inferential component that tests whether a sentiment informed strategy significantly improves risk-adjusted return. Its purpose is to establish whether freely available public sentiment provides an operationally useful, cost-aware early-warning signal for equity volatility.

Interim Project Status (Progress Snapshot)

The current status of the project is summarised below.

- **Completed:** Synopsis submitted; data pipeline designed, implemented, and validated end-to-end by a 27-check automated assertion harness; analysis panel assembled (3,840 ticker-days, eight tickers); data cleaning and exploratory data analysis completed; 24-ticker daily price panel collected (Tiingo); news-sentiment collection complete for all eight sentiment tickers; StockTwits labelled corpus accumulating; literature review completed.
- **In progress:** RQ1 sentiment modelling (FinBERT versus the lexicon baseline); specification of the RQ2 volatility models.
- **Pending:** FinBERT sentiment modelling (RQ1); volatility and direction models (RQ2, RQ3); strategy backtest (RQ4); preliminary results and interpretation.

Scope and Objectives

This study is organised around four research questions, each addressing a distinct aspect of the relationship between retail sentiment and equity risk and drawing on a different family of methods. Each is stated below with its null and alternative hypotheses and the test used to decide between them. All tests are evaluated at a 5% significance level.

Research Question 1 (RQ1): Sentiment Measurement

How accurately can a transformer model (FinBERT) classify StockTwits message sentiment, benchmarked against users' self-tagged bullish and bearish labels and a lexicon baseline? The lexicon baseline combines VADER, a general social-media dictionary, with the Loughran-McDonald finance-specific dictionary (Loughran & McDonald, 2011); the transformer is FinBERT (Araci, 2019). Because 82.4% of the labelled corpus is tagged bullish, accuracy alone is uninformative and the majority-class rate is reported as a floor alongside balanced accuracy, macro-averaged F1 and the area under the receiver operating characteristic curve (ROC-AUC). H0: the accuracy of FinBERT is less than or equal to that of the lexicon baseline. H1: the accuracy of FinBERT exceeds that of the lexicon baseline. Because both classifiers are evaluated on identical messages, the comparison uses McNemar's test for paired nominal data.

Research Question 2 (RQ2): Volatility Prediction

Does aggregated daily sentiment at day t predict next-day realized volatility at t + 1, controlling for lagged volatility and trading volume? A heterogeneous autoregressive (HAR) baseline using the daily, weekly and monthly volatility components is compared with the same specification augmented by the sentiment and attention features, with ticker fixed effects in both, for the reason established in Figure 4. H0: the sentiment coefficients are jointly zero. H1: at least one sentiment coefficient is non-zero. The in-sample comparison uses an incremental F-test and the change in adjusted coefficient of determination; the out-of-sample comparison of forecast accuracy uses the Diebold-Mariano test (Diebold & Mariano, 1995), applied to QLIKE losses following Patton (2011).

Research Question 3 (RQ3): Direction Classification

Do sentiment features improve next-day return-direction prediction beyond a technicals-only baseline? Logistic regression and gradient boosting are each trained on market controls alone and then on the same controls plus sentiment. H0: the sentiment-augmented classifier is no more accurate than the baseline. H1: the augmented classifier is more accurate. As in RQ1 the two models are evaluated on identical test observations, so McNemar's test is again appropriate. The base rate for daily direction is close to 50%, so a null result is an expected and reportable outcome rather than a failure of the design.

Research Question 4 (RQ4): Strategy Evaluation

Does a sentiment-informed trading strategy produce a statistically significant improvement in risk-adjusted return over buy-and-hold, net of transaction costs? The volatility forecast is converted into a transparent exposure rule and back-tested with realistic costs. H0: the Sharpe ratio of the strategy is less than or equal to that of buy-and-hold. H1: the strategy Sharpe ratio is greater. Significance uses the Ledoit-Wolf (2008) robust test for the difference between two Sharpe ratios, which does not assume normally distributed or serially independent returns, implemented with a circular block bootstrap so that volatility clustering is preserved in the resampling.

Literature Survey

Literature Review Approach

Sources were identified through Google Scholar, Scopus, Web of Science, ScienceDirect, SSRN, and arXiv, using Boolean combinations of the study's concept clusters (for example, “investor sentiment” OR “StockTwits” AND “realized volatility” AND “FinBERT” OR “transformer”). Inclusion was limited to peer-reviewed empirical work linking textual sentiment to a measurable market outcome, with field-defining preprints retained where relevant, and priority given to highly cited papers in leading finance and forecasting journals balanced against recency to capture the large-language-model era. Table 1 summarises the most relevant studies.

Table 1

Literature Relevance Matrix

Study	Purpose / Context	Method	Key Findings	Relevance to This Study
Antweiler & Frank (2004)	Message-board sentiment and market activity	Computational-linguistics bullishness index	Messages predict volatility; disagreement predicts volume; return effect small	Anchors the volatility-over-returns focus and the article-count attention feature
Das & Chen (2007)	Extracting sentiment from web small talk	Multi-classifier voting	Sentiment tracks index moves; activity relates to volatility	Early evidence linking posting activity to volatility
Tetlock (2007)	Media tone and the	Media-pessimism	Pessimism predicts price	Frames the

	stock market	factor; VAR	pressure then reversion; extremes predict volume	endogeneity/simultaneity concern
Bollen et al. (2011)	Social-media mood and the market	Mood dimensions + causality tests	Reported mood relates to index moves; noted as fragile	Motivates a reproducible, firm-level design
Loughran & McDonald (2011)	Sentiment in financial text	Finance-specific dictionaries	General dictionaries misclassify financial tone	Basis for the lexicon baseline in RQ1
Sprenger et al. (2014)	Microblog sentiment and trading	ML sentiment + regression	Sentiment relates to returns; disagreement to volatility	Supports microblog sentiment and attention features
Oliveira et al. (2017)	Microblog data for market prediction	Sentiment indicators + multiple ML	Improves volatility/volume forecasts; weaker for returns	Validates the feature family and volatility target
Renault (2017)	StockTwits investor sentiment	Custom lexicon vs. ML	Sentiment predicts intraday index returns; lexicon competitive	StockTwits-specific precedent and baseline
Behrendt & Schmidt (2018)	Twitter sentiment and intraday volatility	GARCH + intraday sentiment	Adds little to intraday volatility once controlled	Cautionary; motivates conservative effect sizes
Araci (2019)	Transformer sentiment for finance	FinBERT (fine-tuned BERT)	Improves financial sentiment classification	The deep-learning method used in RQ1
Audrino et al. (2020)	Sentiment/attention and volatility	Penalized regression on HAR	Attention and StockTwits volume improve realized-volatility forecasts	Closest precedent; frames the per-ticker, business-evaluated gap
Lopez-Lira & Tang (2023)	LLMs and return predictability	GPT-4 on news headlines	Predicts reactions; edges decay as adoption spreads	Frontier LLM evidence; reinforces volatility focus
Yu et al. (2023)	LLM trading agents	FinMem agent with layered memory	Autonomous agent enhances trading decisions	Grounds the optional agentic briefing extension

Three themes organise this literature and connect it directly to the present study. First, on outcome selection, the evidence consistently shows that sentiment and attention forecast volatility and trading volume more reliably than returns, which are closer to efficient (Antweiler & Frank, 2004; Oliveira et al., 2017; Audrino et al., 2020); this study therefore targets next-day volatility as its primary outcome. Second, on sentiment measurement, the field has evolved from finance-specific dictionaries (Loughran & McDonald, 2011) through transformer models (Araci, 2019) to general-purpose large language models (Lopez-Lira & Tang, 2023); RQ1 positions FinBERT against a lexicon baseline within this progression. Third, on the research gap, prior work concentrates at the market-index level, uses lexicon or classical methods, and reports statistical rather than economic value, with limited reproducibility; this study addresses that gap by operationalising transformer sentiment as a per-ticker, next-day volatility early-warning signal, validated through a cost-aware backtest and an open, reproducible pipeline.

Data Description

Data Source(s) and Access

The study draws on three free, publicly accessible sources. Daily split and dividend-adjusted price data are obtained from Tiingo (https://www.tiingo.com); historical per-ticker news sentiment from Alpha Vantage (https://www.alphavantage.co); and recent labelled messages from the StockTwits API (https://api.stocktwits.com). Prices supply the target and market controls, Alpha Vantage supplies the historical sentiment feature for RQ2 to RQ4, and StockTwits supplies the human-labelled corpus for RQ1. During development, an initial price source (Stooq) was found to block automated access and Alpha Vantage's full-history price endpoint was available at a premium price, so Tiingo was adopted after validating access. This deliberate evaluation of data sources strengthens reproducibility and is noted as a practical constraint in the limitations of the final report.

Dataset Overview

The assembled analysis panel is organised as a ticker-day table combining price-derived controls, the next-day realized-volatility target, and news-sentiment features.

- Number of records (rows): 3,840 usable ticker-days (eight sentiment tickers, 480 trading days each) drawn from a 24-ticker, 11,873-row price panel.
- Number of variables (columns): 15 principal variables (see Table 2).
- Time period: 3 January 2023 to 31 December 2024.
- Unit of analysis: the ticker-day.

- Target variable(s): next-day realized volatility (rv_next), estimated from daily open-high-low-close (OHLC) prices via the Parkinson estimator.

Data Dictionary (Mandatory)

Table 2

Data Dictionary (Principal Variables)

Variable Name	Source	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
symbol	All	Stock ticker symbol identifying the equity	string	24 tickers (see config.TICKERS)	None permitted; row invalid without it	Key. Analysis sample restricted to the 8 tickers in NEWS_TICKERS
date	All	Trading date (US/Eastern) of the observation	date	2023-01-03 to 2024-12-31, NYSE trading days	None permitted; row invalid without it	Key. Non-trading days absent by construction, not imputed
close	Tiingo	Split- and dividend-adjusted daily closing price	float	> 0 USD	None observed	Raw. Adjusted series, so corporate actions create no false volatility
log_return	Derived	Daily log return, ln(close_t / close_t-1)	float	approx. -0.5 to +0.5 log points	Null on each ticker's first day; row excluded	Control. Used lagged, never contemporaneously with the target
abs_return	Derived	Absolute daily log return	float	>= 0 log points	Null where log_return is null; row excluded	Control. Correlated with rv; monitored via VIF (Table 5)
log_volume	Derived	Log adjusted share volume, log(1 + volume)	float	> 0 ln(shares)	None observed	Control. log1p so a zero-volume day maps to 0, not minus infinity
rv	Derived	Realized volatility at t from the high-low range (Parkinson)	float	> 0 daily sigma	None observed	Control. Known at the close of t; daily HAR component
rv_lag1	Derived	Realized volatility at t-1	float	> 0 daily sigma	Null on each ticker's first day; row excluded	Control. Retained for continuity with the synopsis specification
rv_w	Derived	Weekly HAR component: mean rv over t-4 to t inclusive	float	> 0 daily sigma	Null for the first 4 days per ticker; rows excluded	Control. Window ends at t because rv_t is observable at that close
rv_m	Derived	Monthly HAR component: mean rv over t-21 to t inclusive	float	> 0 daily sigma	Null for the first 21 days per ticker; rows excluded	Control. Binding on sample size: costs 21 ticker-days per ticker
rv_next	Derived	Next-day realized volatility (t+1)	float	> 0 daily sigma	Null on each ticker's final day; row excluded	TARGET. The only column using information dated after t, by design
news_sent_wmean	Alpha Vantage	Relevance-weighted mean news sentiment on day t	float	-1 (bearish) to +1 (bullish)	Set to 0.0 (neutral) on queried tickers with no articles, flagged by news_missing; null for tickers never queried	Predictor. Weights are Alpha Vantage relevance scores
news_article_count	Alpha Vantage	Articles referencing the ticker on day t; attention proxy	int	>= 0 articles	Set to 0 on queried tickers with no articles (a genuine low-attention observation); null for tickers never queried	Predictor. Zero-article days retained; dropping them truncates attention at 1

Variable Name	Source	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
news_missing	Derived	1 if the ticker-day had no articles, 0 otherwise	int	{0, 1}	Null for tickers never queried for news	Flag. Separates neutral coverage from absent coverage
log_articles	Derived	Log attention proxy, log(1 + news_article_count)	float	>= 0	Null where news_article_count is null	Predictor. Well defined only because zero-article days are retained

Note. The panel contains 15 variables. The StockTwits labelled corpus (id, created_at, symbol, body, sentiment_basic) is used separately for RQ1 sentiment-model validation and is not part of the historical RQ2–RQ4 panel; no user-identifying fields are retained. Generated by src/make_data_dictionary.py and versioned at data/data_dictionary.md.

GitHub Data Availability Statement

The raw and processed datasets and all code are openly available in the project repository:

- Raw data: data/raw/{prices, news, stocktwits}/
- Processed/clean data: data/processed/panel.csv
- Code/notebooks: src/ (collection, feature building, and dictionary scripts)

Screenshots / Evidence

[Insert Figures: repository folder tree, and dataset preview screenshots of the raw folders and the assembled panel.csv, each labelled and referenced in the text (carried over from the synopsis figures).]

Analysis

This section documents how the analysis sample was constructed from the collected data, then reports the exploratory analysis that informed the modelling decisions described in the following section. Every figure and table reported here is generated by src/run_eda.py in the project repository, so all values are reproducible from the committed data.

Data Cleaning

Cleaning is implemented in src/build_features.py and verified by an automated assertion harness, src/validate_panel.py. Rather than inspecting the output by eye, the harness recomputes every derived variable directly from the raw source files and fails if any value disagrees. It runs 27 checks spanning key uniqueness, definitional consistency, news integrity, corporate actions, and the point-in-time test described at the end of this subsection. All 27 pass on the current panel, and the script exits with an error status if any fails, so a corrupted panel cannot silently reach the modelling stage.

The collected price panel contains 11,873 ticker-days across 24 technology equities. This is 175 fewer than the 12,048 implied by 24 tickers over 502 trading days, because Arm Holdings (ARM) listed in September 2023 and has no earlier history. Restricting to the eight tickers for which news sentiment was collected leaves 4,016 ticker-days. Removing the final trading day of each series, where the next-day target is undefined, and the first 21 days, where the monthly heterogeneous autoregressive (HAR) component has insufficient history, yields a balanced analysis sample of 3,840 ticker-days, exactly 480 per ticker. Table 3 records each step with counts and percentages.

Table 3

Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Records Affected	Rationale
Duplicate ticker-days	symbol, date	Key-uniqueness assertion in validate_panel.py	None found; no action required	0 (0.00%)	Data quality
Tickers outside the sentiment universe	news_sent_wmean, news_article_count	Free-tier quota of 25 requests per day	Excluded from the analysis sample; retained in the price panel as market context	7,857 (66.18%)	Quota constraint, not a data defect
Target undefined at series end	rv_next	Null check after the t to t+1 shift	Excluded, one ticker-day per ticker	8 (0.07%)	Next-day volatility does not exist for the final

					observation
HAR burn-in window	rv_m	22-day rolling mean requires 22 prior observations	Excluded, 21 ticker-days per ticker	168 (1.42%)	Monthly HAR component undefined until the window fills
No news articles on a trading day	news_sent_wmean, news_article_count	Gap between the news files and the trading calendar	Retained as zero-attention observations: article count 0, sentiment 0 (neutral), indicator news_missing = 1	945 (7.96%)	A zero-article day is a genuine low-attention observation, not a missing measurement
Extreme one-day price moves	close, rv	Screen for absolute close-to-close moves above 30%, cross-checked against split ratios and the volatility path	Both flagged moves verified as post-earnings repricings and retained; no winsorising or trimming applied	2 (0.02%)	Extreme volatility is the phenomenon under study, not noise to remove
User-identifying fields	user_id, user_followers	Privacy review of the StockTwits collector	Removed at collection and stripped from 23 existing files	23 files	Ethics: no individual user is profiled or re-published

Note. Percentages are of the 11,873-row collected price panel. Exclusions are sequential, leaving an analysis sample of 3,840 ticker-days. Counts are produced by src/run_eda.py and written to docs/tables/eda_cleaning_log.csv.

Two treatment decisions warrant explanation. The first concerns the 945 ticker-days, 24.6% of the analysis sample, on which no news article mentioned the ticker. An earlier version of the pipeline discarded these rows, because joining the news files to the trading calendar produced no match. That treatment was incorrect. A day on which no article mentions a ticker is a genuine low-attention observation rather than a failed measurement, and discarding such days truncates the attention variable at a minimum of one article, removing exactly the low-attention contrast the study depends upon. The loss would also have fallen unevenly: Palantir (PLTR) would have forfeited 73% of its history while Microsoft (MSFT) forfeited 4%, leaving a severely unbalanced panel. These days are therefore retained with an article count of zero, a neutral sentiment score, and an indicator variable that allows the models to distinguish neutral coverage from absent coverage. The correction increased the analysis sample from 3,026 to 3,840 ticker-days and produced a balanced panel of 480 days per ticker.

This treatment is applied only to the eight tickers actually queried. For the remaining sixteen no request was ever issued, so their news fields remain null; recording a zero there would fabricate an observation from an absence of collection. The validation harness asserts that no such value exists.

The second decision concerns outliers. No trimming or winsorising was applied. A screen for absolute close-to-close moves above 30% flagged two observations: ARM on 8 February 2024 (+47.9%) and PLTR on 6 February 2024 (+30.8%). Each was tested in two ways. An unadjusted stock split produces a price ratio sitting almost exactly on a simple fraction, and neither ratio, 1.479 and 1.308, does so. A split also leaves the price discontinuous while realized volatility is unchanged, whereas both observations show volatility rising sharply over several sessions, ARM from 0.030 to 0.179. Both are therefore post-earnings repricings rather than data errors. Since the object of the study is to forecast exactly such episodes, removing them would remove the signal of interest.

Because the study makes a forecasting rather than an explanatory claim, cleaning must guarantee that no predictor contains information dated after day t. This is verified rather than asserted. For a random sample of ticker-days the harness rebuilds every predictor from raw records restricted to dates on or before t and requires an exact match with the stored value; across all 11,873 ticker-days the largest discrepancy is of the order of ten to the power of minus sixteen, that is, floating-point rounding. Realized volatility at t is computed from that day's high and low prices, so it is known at the close of day t and is admissible as a predictor of day t + 1. The HAR windows accordingly end at t rather than being shifted back, which is the standard specification and introduces no look-ahead bias.

Exploratory Data Analysis

Table 4 summarises the analysis sample. The target and its lagged components share a common scale, so the differences between them are informative: averaging volatility over longer windows progressively removes the right tail, with skewness falling from 1.81 for the daily series to 0.62 for the monthly average.

Table 4

Descriptive Statistics for the Analysis Sample

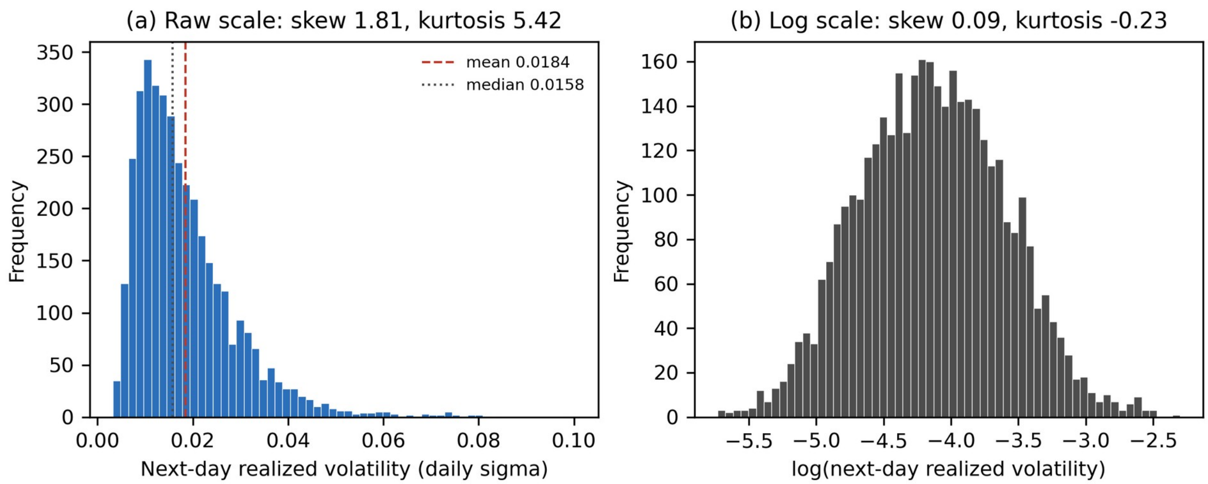
Variable	N	M	Mdn	SD	IQR	Min	Max	Skew	Kurtosis
rv_next (target)	3,840	0.0184	0.0158	0.0107	0.0124	0.0032	0.1003	1.81	5.42
rv (daily HAR)	3,840	0.0185	0.0158	0.0108	0.0124	0.0032	0.1003	1.81	5.35
rv_w (weekly HAR)	3,840	0.0185	0.0170	0.0085	0.0117	0.0045	0.0583	1.07	1.43
rv_m (monthly HAR)	3,840	0.0186	0.0178	0.0074	0.0115	0.0070	0.0467	0.62	-0.11
news_sent_wmean	3,840	0.1365	0.1320	0.1774	0.2677	-0.6165	0.8746	-0.12	0.57
news_article_count	3,840	2.20	2.00	2.32	2.00	0	26	2.30	12.10
log_articles	3,840	0.9354	1.0986	0.6770	0.6931	0.0000	3.2958	0.08	-0.82
log_return	3,840	0.0022	0.0013	0.0275	0.0252	-0.1638	0.2685	1.08	10.54
log_volume	3,840	17.85	17.74	0.99	1.17	15.37	21.16	0.63	0.10

Note. Realized volatility is expressed in daily sigma units. Sentiment is relevance-weighted and bounded at -1 and +1. Values are written to docs/tables/eda_descriptives.csv.

Figure 1 shows why the target is modelled on a logarithmic scale. On the raw scale next-day realized volatility is strongly right-skewed, with a skewness of 1.81 and a kurtosis of 5.42: most days are quiet and a small number are extreme. Taking logarithms reduces these to 0.09 and -0.23 respectively, which is close to normal. Two modelling decisions follow. The regression target is the logarithm of next-day realized volatility, so that estimates are not dominated by a handful of earnings days, and forecast accuracy is reported using QLIKE alongside RMSE, because squared-error loss on a fat-tailed target rewards a model that fits the extreme tail at the expense of the remaining observations.

Figure 1

Distribution of Next-Day Realized Volatility, Raw and Log Scale

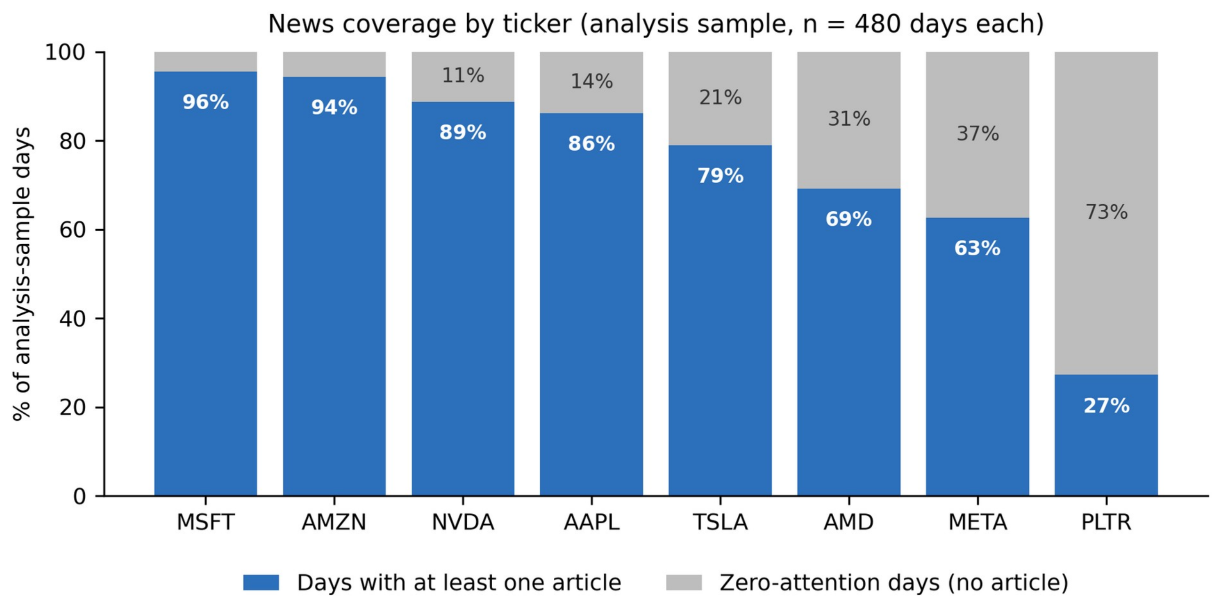


Note. Panel (a) shows the raw target and panel (b) its natural logarithm, each with 60 bins.

Figure 2 shows that news coverage is highly uneven across the sentiment universe. Microsoft is covered on 96% of trading days and Amazon on 94%, whereas Palantir is covered on only 27%. Coverage is therefore not an incidental property of the data but a characteristic that differs systematically between firms, which is the principal justification for retaining zero-attention days and including the news_missing indicator. It also anticipates the confounding problem identified in Figure 5.

Figure 2

News Coverage by Ticker Within the Analysis Sample

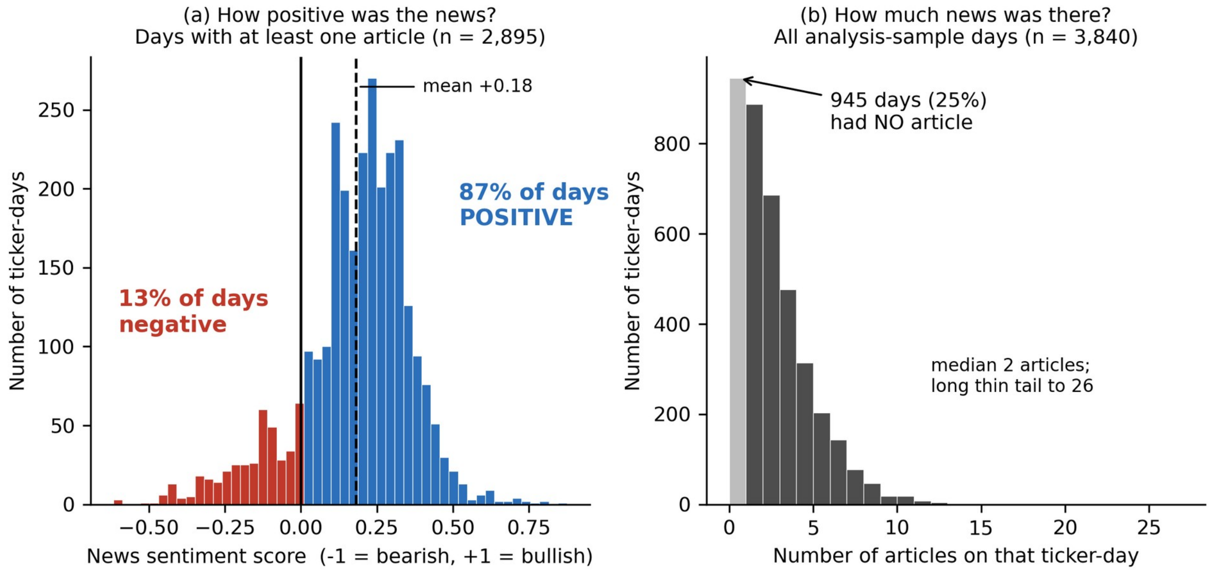


Note. Each ticker contributes exactly 480 ticker-days. Shaded portions are days on which no article mentioned the ticker.

Figure 3 reveals an asymmetry with direct consequences for the study. Relevance-weighted sentiment averages +0.181 and is positive on 87% of days with news, so the measure carries considerably less negative than positive variation. This compresses the range over which any sentiment effect could be detected and is recorded as a limitation. Panel (b) shows that attention is heavily concentrated at low values, with a median of two articles and a maximum of 26, which is why it enters the models in logarithmic form.

Figure 3

Distribution of News Sentiment and Attention

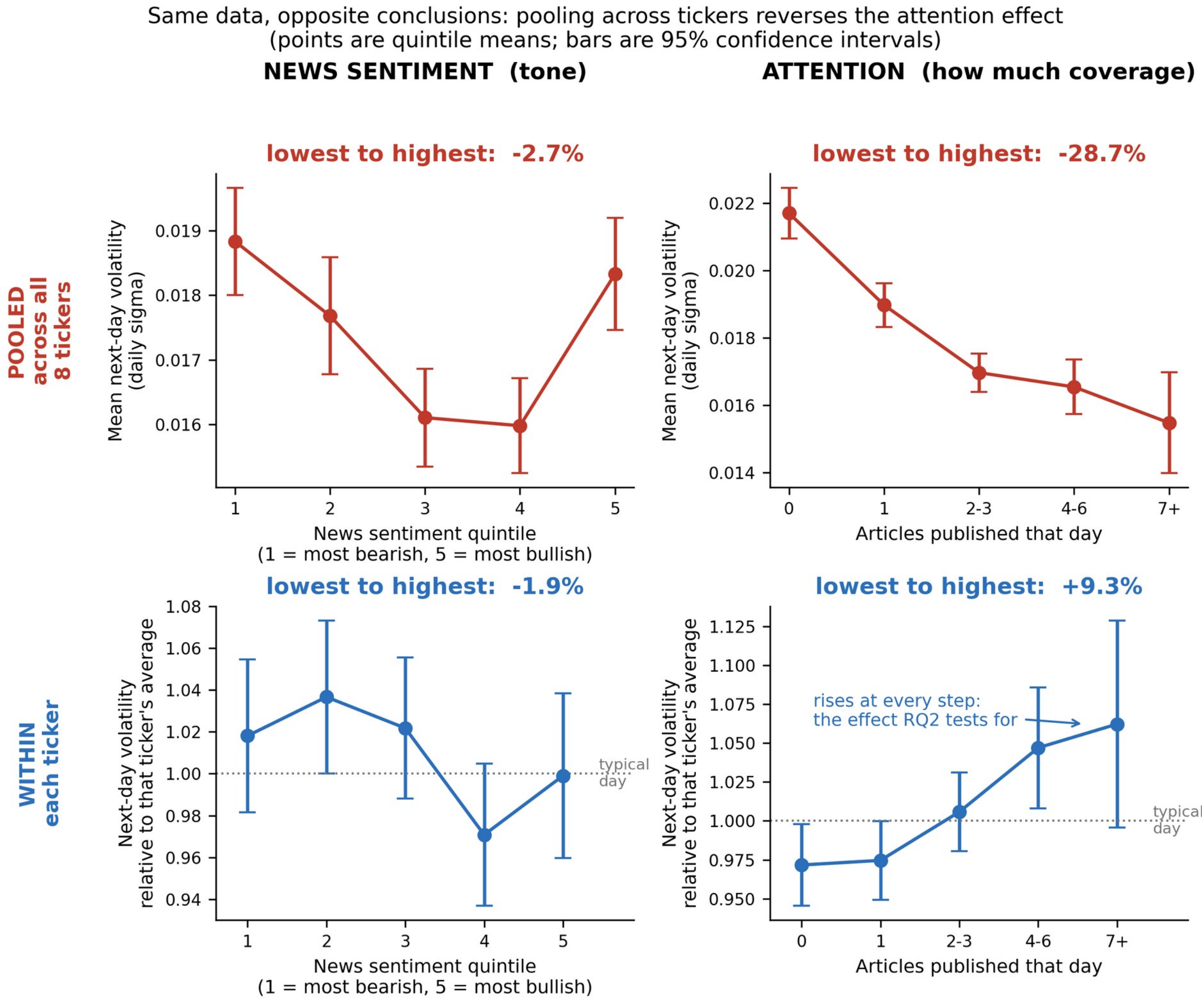


Note. Panel (a) is restricted to days with at least one article; panel (b) includes all analysis-sample days.

Figure 4 contains the most consequential finding of the exploratory analysis. Pooled across tickers, moving from the lowest to the highest attention band is associated with 28.7% lower next-day volatility, which contradicts the established result that attention and volatility move together (Audrino et al., 2020). Comparing each ticker against its own mean reverses the sign: within tickers, days with seven or more articles carry 9.3% higher next-day volatility than days with none, and the relationship rises monotonically across all five bands. Sentiment behaves differently, showing no consistent pattern in either specification, at -2.7% pooled and -1.9% within ticker. The pooled estimate is an artefact, for the reason set out in Figure 5, and the practical consequence is that the models addressing RQ2 must include ticker fixed effects. Attention, rather than sentiment tone, emerges as the stronger candidate predictor.

Figure 4

Next-Day Volatility by Predictor Group, Pooled and Within Ticker

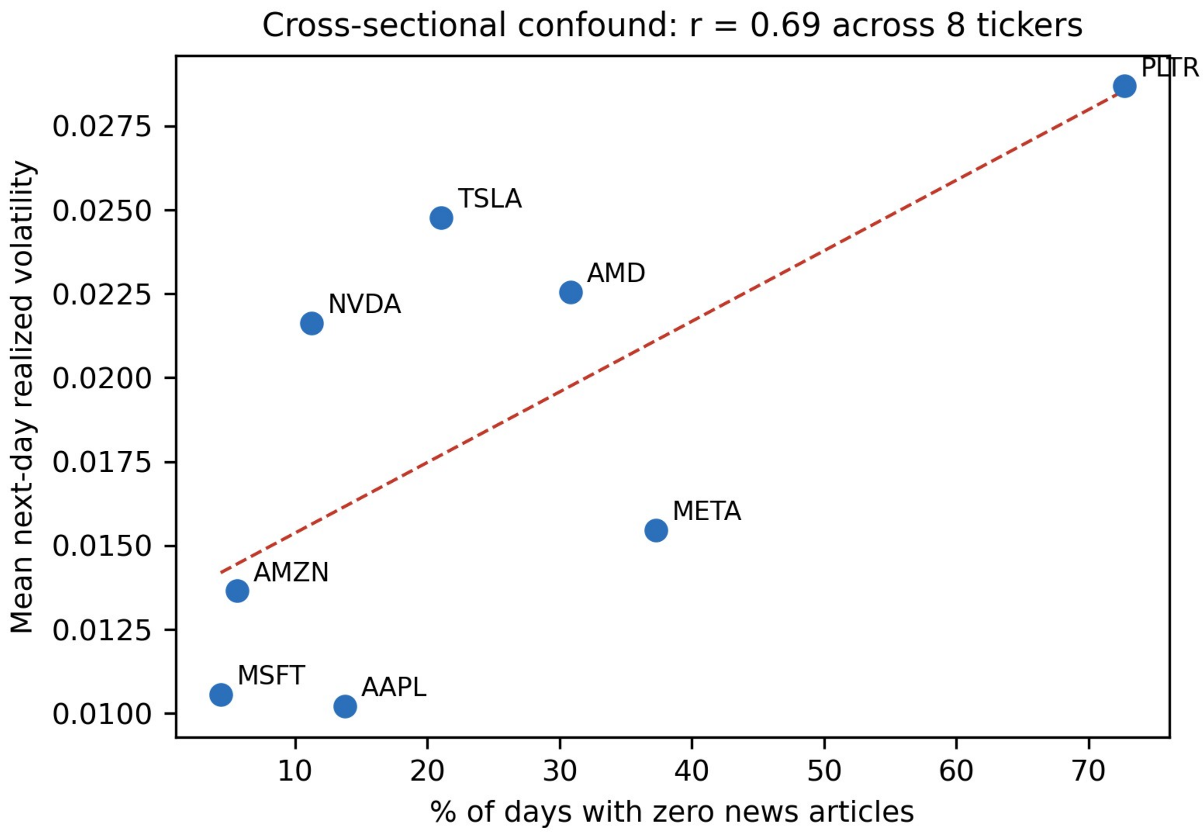


Note. Upper panels pool all tickers; lower panels express the target relative to each ticker's own mean, so that 1.00 denotes a typical day for that stock. Sentiment is grouped into quintiles. Attention is grouped by article count rather than quintiles, because a quarter of ticker-days have exactly zero articles and quantile boundaries would either collapse or split identical observations. Error bars are 95% confidence intervals.

Figure 5 explains the sign reversal. Across the eight tickers, the proportion of zero-attention days correlates 0.69 with average volatility. Palantir has the least coverage, with no article on 73% of days, and the highest average volatility at 0.0287, while Microsoft has the most coverage, with no article on only 4% of days, and among the lowest volatility at 0.0106. In pooled data, low attention therefore functions largely as a label for which stock is being observed rather than as information about what is about to happen to it. This is a Simpson's paradox: a relationship that holds within each group reverses when the groups are combined. Because RQ2 asks whether a change in a ticker's own attention predicts a change in its own next-day volatility, only the within-ticker comparison is informative, and pooled results are reported solely as a cautionary contrast.

Figure 5

Cross-Sectional Confounding Between Coverage and Baseline Volatility

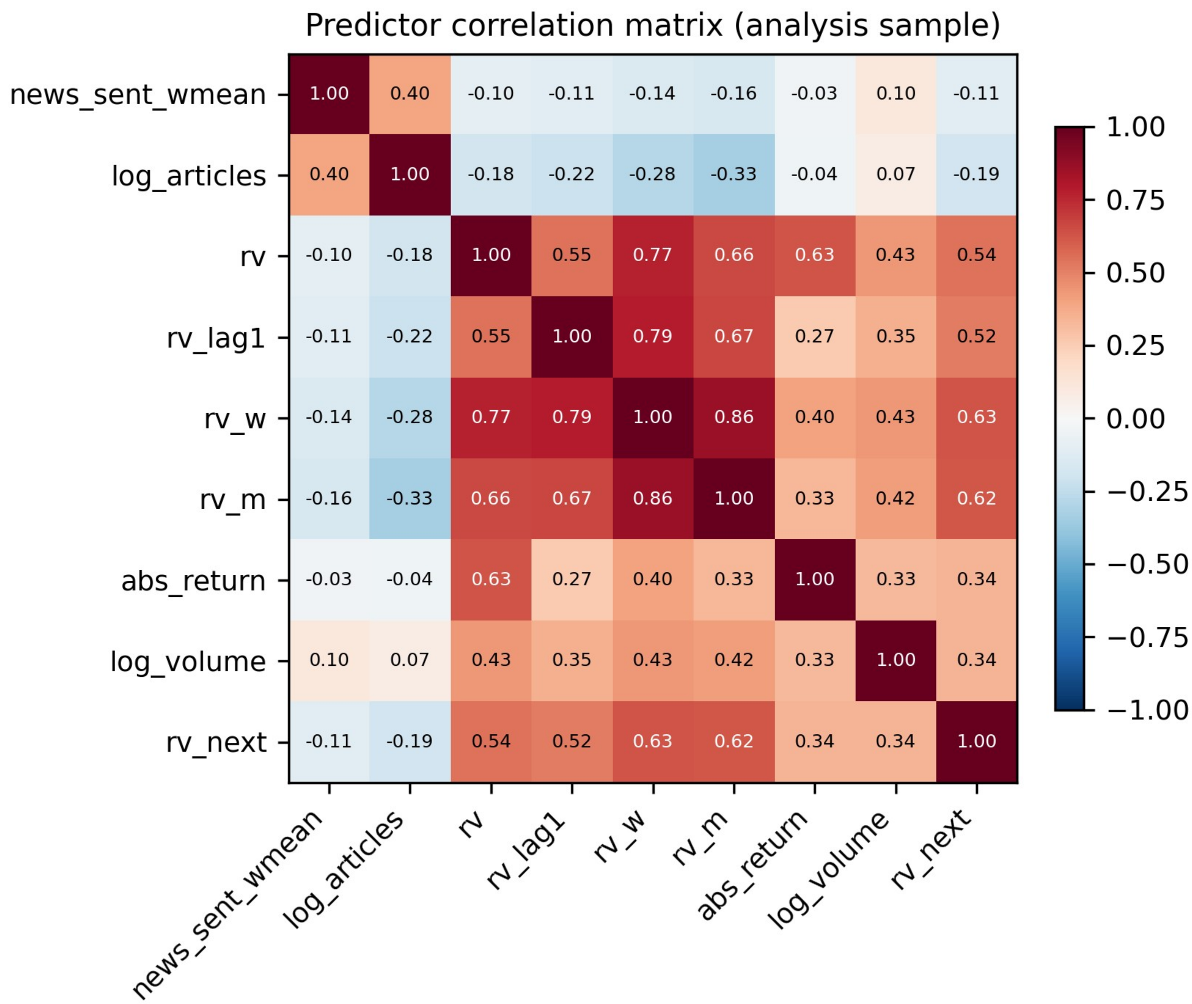


Note. Each point is one of the eight tickers in the analysis sample; the dashed line is an ordinary least squares fit.

Figure 6 confirms that lagged volatility dominates the predictor set. The weekly HAR component correlates 0.63 with the target, while news sentiment correlates only -0.11 and attention -0.15. This is the expected ordering and it determines how RQ2 must be evaluated. Sentiment cannot be judged by its raw association with next-day volatility, which is small, but by whether it adds explanatory power once the HAR components are already in the model. The hypothesis test for RQ2 is therefore an incremental one.

Figure 6

Correlation Matrix of Predictors and Target



Note. Pearson correlations computed on the 3,840-observation analysis sample.

Because several controls measure closely related quantities, multicollinearity was assessed using variance inflation factors, reported in Table 5. Step 4 of the research design identifies a factor above 10 as indicating severe multicollinearity.

Table 5

Variance Inflation Factors for the Full Predictor Set

Predictor	VIF	Assessment
rv_w (weekly HAR component)	7.91	Moderate (5-10)
rv_m (monthly HAR component)	4.20	Acceptable (below 5)
rv (daily HAR component)	3.63	Acceptable (below 5)
rv_lag1	2.71	Acceptable (below 5)
abs_return	1.74	Acceptable (below 5)
log_volume	1.40	Acceptable (below 5)
log_articles	1.36	Acceptable (below 5)
news_sent_wmean	1.20	Acceptable (below 5)

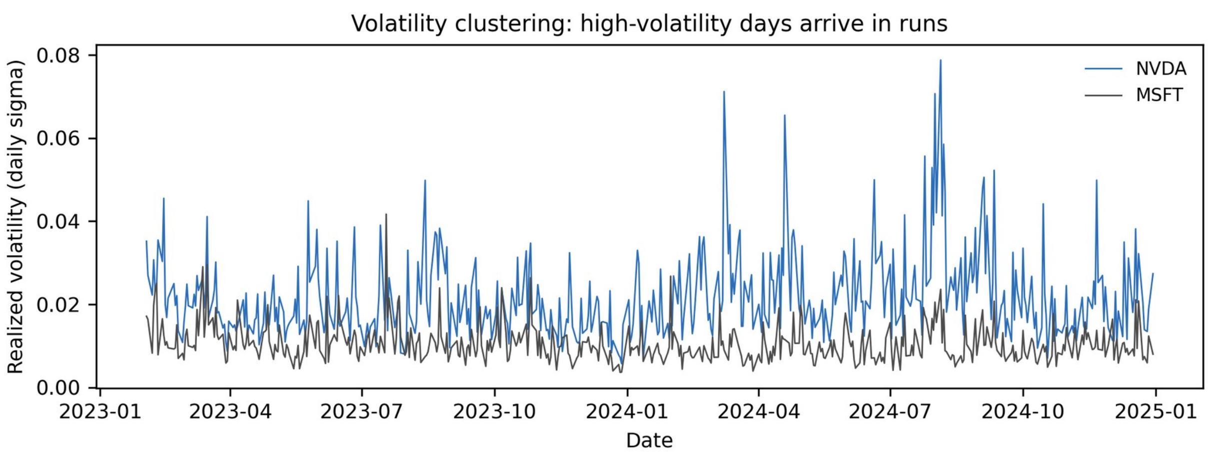
Note. Computed on the analysis sample and written to docs/tables/eda_vif.csv. No predictor exceeds the severe threshold of 10.

No predictor exceeds the severe threshold. The weekly HAR component is the most collinear at 7.91, which is expected because it is a moving average of the daily series, and it remains within the range normally considered tolerable. The two sentiment variables are close to orthogonal to the market controls, at 1.36 and 1.20, which is an encouraging result for RQ2: whatever incremental contribution they make will not be an artefact of overlap with lagged volatility. The full control set is therefore retained without modification.

Finally, Figure 7 illustrates the volatility clustering that motivates the HAR specification.

Figure 7

Realized Volatility Over Time for NVDA and MSFT



Note. Daily realized volatility across the 2023 to 2024 study window for two contrasting large-capitalisation tickers.

Volatility arrives in runs rather than independently from day to day: quiet periods persist and turbulent periods persist. This is why a forecasting model can achieve substantial accuracy from lagged volatility alone, and why the appropriate benchmark for RQ2 is a HAR baseline rather than a naive mean. It also sets a demanding standard for the sentiment features, which must improve upon a baseline that is already informative.

Table 6

Exploratory Data Analysis Insight Summary

Figure/Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ/Objective)	Decision/Next Step
Figure 1	Distribution of the target, raw and logged	Right-skewed (1.81), fat-tailed; near-normal once logged	RQ2 estimates would be dominated by a few extreme days	Model the log target; report QLIKE alongside RMSE
Figure 2	News coverage by ticker	Coverage ranges from 27% (PLTR) to 96% (MSFT)	Determines whether zero-attention days can be dropped	Retain zero-attention days; include news_missing
Figure 3	Sentiment and attention distributions	Sentiment positive on 87% of days; attention concentrated at low counts	Limited negative variation caps the detectable sentiment effect	Use log attention; record the asymmetry as a limitation
Figure 4	Group means, pooled versus within ticker	Attention effect reverses sign: -28.7% pooled, +9.3% within	A pooled model would report the wrong sign for RQ2	Include ticker fixed effects in the RQ2 and RQ3 models
Figure 5	Coverage against baseline volatility across tickers	Correlation of 0.69 produces a Simpson's paradox	Identifies the cause of the reversal in Figure 4	Report within-ticker results as primary, pooled as contrast
Figure 6	Predictor correlation matrix	rv_w correlates 0.63 with the target; sentiment only -0.11	Sentiment cannot be assessed on raw correlation alone	Test sentiment by incremental contribution over HAR
Table 5	Variance inflation factors	Maximum VIF 7.91; sentiment features at 1.36 and 1.20	Confirms the coefficients will be interpretable	Retain the full control set unchanged
Figure 7	Realized volatility over time	Pronounced volatility clustering	Justifies the HAR benchmark used for RQ2	Benchmark against HAR rather than a naive mean

Note. Each row states the interpretation drawn from the exhibit and the design decision that follows from it.

Sample Size Computation

Each research question is sized by the method matching its statistical form: a confidence interval where a single quantity is estimated, and power analysis where an effect is tested. Effect sizes are set conservatively, because the literature consistently reports that sentiment effects on market outcomes are statistically real but economically small (Antweiler & Frank, 2004; Audrino et al., 2020). Table 7 states the method, parameters and substitution for each question.

Table 7

Minimum Sample Size by Research Question

RQ	Method	Formula	Parameters	Substitution	Minimum N
RQ1	Confidence interval for a proportion	n = Z^2 p(1-p) / e^2	Z = 1.96 (95%); p = 0.50; e = 0.05	(1.96)^2 (0.50)(0.50) / (0.05)^2	385 labelled messages
RQ2	Power analysis, multiple regression (Cohen, 1988)	N = L / f^2 + u + 1	alpha = .05; power = .80; f^2 = 0.02 (small); u = 3; L = 10.90	10.90 / 0.02 + 3 + 1	549 ticker-days
RQ3	Power analysis, one-sample proportion	n = [Z(1-a) sqrt(p0 q0) + Z(1-b) sqrt(p1 q1)]^2 / (p1 - p0)^2	Z(1-a) = 1.645; Z(1-b) = 0.8416; p0 = 0.52; p1 = 0.57	[1.645 sqrt(.2496) + .8416 sqrt(.2451)]^2 / (.05)^2	614 ticker-days
RQ4	Power analysis, one-sample t test on the Sharpe ratio	n = [Z(1-a) + Z(1-b)]^2 / d^2	Z(1-a) = 1.645; Z(1-b) = 0.8416; d = 0.063 (annual Sharpe 1.0 / sqrt(252))	(1.645 + 0.8416)^2 / (0.063)^2	1,558 trading days

Note. Effect sizes follow Cohen's (1988) benchmarks. For RQ2 the tested predictors are relevance-weighted sentiment, log attention and the zero-attention indicator, so u = 3 and the corresponding tabulated constant is L = 10.90.

Because all four analyses draw on the same collected data, the study adopts the largest of the four minima. Three of the four requirements are met comfortably: the labelled StockTwits corpus contains 4,222 messages against the 385 required for RQ1, and the analysis panel contains 3,840 ticker-days against the 549 and 614 required for RQ2 and RQ3. RQ4 is the binding and unmet constraint, and is reported honestly rather than absorbed into the pooled figure. Its requirement of 1,558 observations is expressed in trading days, and the study window contains only 500. Pooling the eight tickers to 3,840 ticker-days does not resolve this, because daily returns on large-capitalisation technology stocks are strongly cross-correlated, so the effective number of independent observations is far below the row count. RQ4 is therefore under-powered by design, its test is implemented with a block bootstrap to respect that dependence, and a null result must be interpreted as inconclusive rather than as evidence of no effect.

Meeting a minimum sample size is necessary but not sufficient, because the ticker-days are not independent of one another. Table 8 therefore restates the available sample on three bases and reports the smallest effect RQ2 could detect on each. Even on the most conservative basis the design retains power to detect a small effect in Cohen's (1988) terms, which is the magnitude the literature leads the study to expect.

Table 8

Effective Sample Size and Minimum Detectable Effect for RQ2

Basis	N	Rationale	Detectable f^2	Interpretation
Pooled ticker-days	3,840	Raw row count; assumes every observation is independent	0.003	Optimistic; overstates precision
Correlation-adjusted	1,093	8 / (1 + 7 x 0.359) = 2.28 effective series x 480 dates	0.010	Primary basis reported in this study
Distinct trading dates	480	Treats each date as one cluster, ignoring cross-sectional information	0.023	Most conservative; still near the small-effect threshold

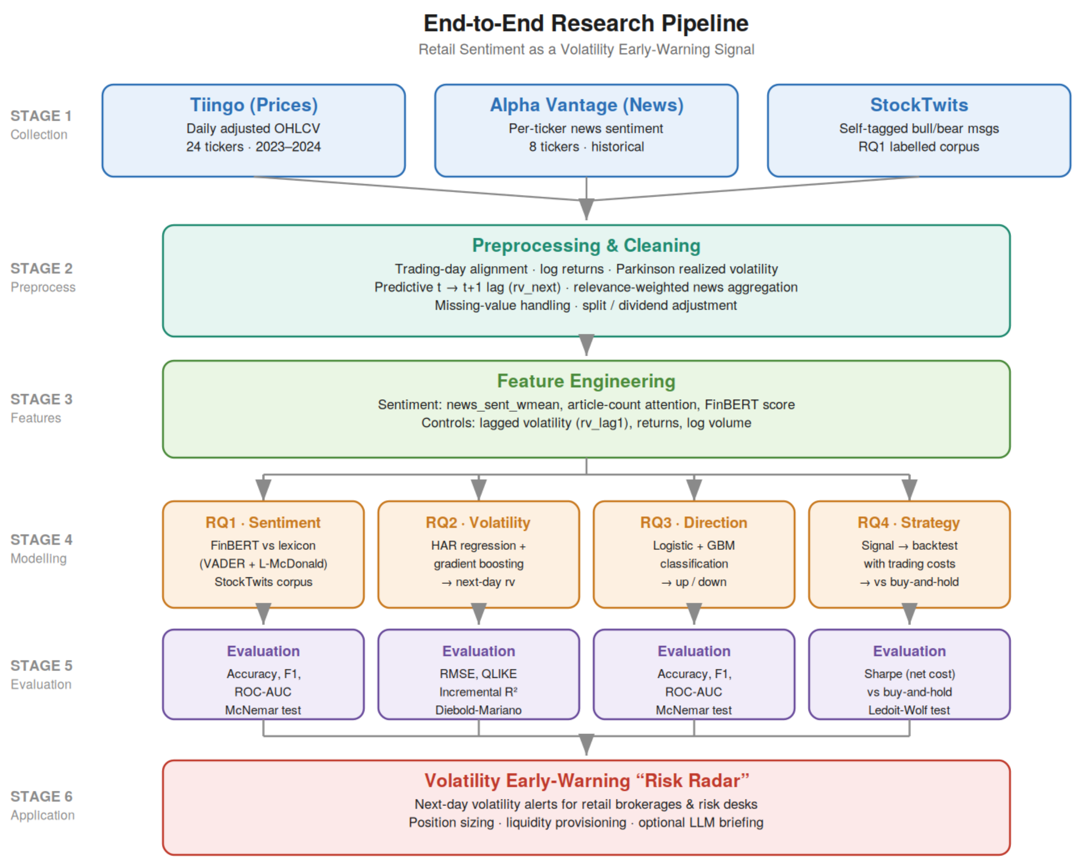
Note. Computed at alpha = .05 and power = .80 with u = 3 tested predictors and L = 10.90 (Cohen, 1988). Cohen's benchmarks are f^2 = 0.02 (small), 0.15 (medium) and 0.35 (large). Generated by src/run_eda.py.

Modelling

The overall research pipeline, from data collection through preprocessing, feature engineering, modelling, and evaluation, is shown in Figure 8.

Figure 8

End-to-End Research Pipeline



Note. The pipeline maps each of the four research questions to its model family and evaluation metrics, with StockTwits feeding RQ1 and Alpha Vantage news feeding RQ2–RQ4.

Data Preprocessing

Preprocessing is deterministic and executed entirely in src/build_features.py, so the panel can be rebuilt from the raw files by a single command. Messages and articles are assigned to United States Eastern trading dates, with items published after the 16:00 close attributed to the following trading day, so that post-close information informs t + 1 rather than t. The predictive alignment is created by shifting realized volatility backwards by one trading day within each ticker, which defines the target while leaving every predictor dated on or before t.

Missing values are handled by category rather than by a single global rule. Structural gaps, namely the first 21 trading days of each series where the monthly HAR window is incomplete and the final day where the target is undefined, are removed rather than imputed, since imputing them would invent history. Zero-article days are not treated as missing at all: they are recorded as genuine low-attention observations, as set out in the data cleaning subsection. No imputation of prices or volumes was required, because the adjusted Tiingo series is complete across the window.

Transformation is dictated by the model. Trading volume and article counts are heavily right-skewed and enter in logarithmic form, and the regression target is log-transformed on the evidence of Figure 1. Predictors are standardised to zero mean and unit variance for logistic regression, where the scale of a coefficient would otherwise depend on the units of its variable; no scaling is applied for gradient boosting, because tree splits are invariant to any monotonic rescaling of a feature. The only categorical variable is the ticker symbol, which is not one-hot encoded as a predictor but absorbed as a fixed effect, which is the appropriate treatment for a panel in which each unit contributes many observations.

Experimental Design and Validation Strategy

Each research question is a controlled comparison. The control condition is a model containing only market information; the experimental condition is the identical model, algorithm and hyperparameters, with sentiment features added. Both are estimated on the same training windows and scored on the same test observations, making the comparison paired and isolating the contribution of sentiment from any difference in model class. Reproducibility is supported by fixed random seeds, a public repository, and the validation harness described earlier.

Models are validated by walk-forward analysis rather than k-fold cross-validation. Although Step 4 of the research design proposes k-fold as the default safeguard against overfitting, random partitioning is not admissible here: assigning observations to folds at random would place future trading days in the training set and past days in the test set, allowing the model to learn from information that would not have been available at the time of the forecast. Walk-forward validation instead trains on an initial contiguous block, tests on the block immediately following it, then advances both windows and repeats, so that every prediction is made using only prior data and the evaluation mirrors live operation. Splits are formed on dates rather than rows, so that all eight tickers move through the windows together and no ticker's future can inform another ticker's past.

Regression Diagnostics and Model Selection

The HAR specification is assessed against the assumptions of linear regression. Residuals are inspected for non-linearity and for heteroscedasticity, which is expected in volatility data and is addressed by reporting standard errors clustered by date, so that inference accounts for the cross-sectional correlation identified in Figure 5. Normality of residuals is examined on the log scale, where Figure 1 shows the target is close to symmetric. Multicollinearity has already been assessed in Table 5, where no variance inflation factor exceeds the threshold of 10 given in Step 4. Where nested specifications are compared, model selection uses the adjusted coefficient of determination together with the Akaike and Bayesian information criteria, which penalise the additional parameters that the sentiment features introduce, so that any reported improvement reflects genuine explanatory gain rather than the mechanical rise in fit that follows from adding predictors.

Choice of Models with Justification

Each research question is framed as a baseline-versus-augmented comparison, so that any improvement is attributable to sentiment rather than to model class. Models were selected on the basis of the literature, the data type, interpretability, and the evaluation task.

Model 1 — FinBERT vs. lexicon (RQ1): a transformer classifier (FinBERT) benchmarked against a VADER + Loughran–McDonald lexicon baseline, graded on StockTwits self-tagged labels. Justified by FinBERT's demonstrated gains in financial sentiment classification (Araci, 2019) and the transparency of the lexicon baseline (Loughran & McDonald, 2011).

Model 2 — HAR regression + gradient boosting (RQ2): a heterogeneous-autoregressive baseline for realized volatility, extended with sentiment features and a gradient-boosting regressor to capture non-linearities. Justified by the HAR model's standing in volatility forecasting and prior sentiment-volatility evidence (Audrino et al., 2020).

Model 3 — Logistic regression + gradient boosting (RQ3): a technicals-only classifier of next-day return direction, compared with a sentiment-augmented version. Justified by interpretability (logistic) and predictive strength (gradient boosting) on tabular data.

Model 4 — Sentiment-informed strategy backtest (RQ4): a transparent trading rule derived from the volatility signal, back-tested with transaction costs against buy-and-hold. Justified by the need to demonstrate economic, cost-aware value rather than statistical significance alone.

Features Included and Feature Engineering

The feature set combines news-derived sentiment features with lagged market controls; the target is the next-day realized volatility. Table 9 summarises the feature set and its usage.

Table 9

Feature Set and Usage

Feature	Type	Engineering	Used in
news_sent_wmean	Sentiment	Relevance-weighted daily mean	RQ2, RQ3, RQ4
news_article_count	Attention	Daily article count (log)	RQ2, RQ3
FinBERT sentiment	Sentiment (DL)	RQ1 model applied to news text	RQ2, RQ3
rv_lag1	Control	Prior-day realized volatility	RQ2

log_return, abs_return	Control	Daily returns	RQ2, RQ3
log_volume	Control	log(1 + volume)	RQ2, RQ3

Evaluation Metrics (Formulae and Calculations)

Metrics are matched to each question's form. Table 10 lists the metrics, their formulae, and the research question to which each applies; worked illustrative calculations are retained from the synopsis and will be recomputed on real results in the final report.

Table 10

Evaluation Metrics and Formulae

Metric	Formula	Applies to
Accuracy	(TP + TN) / (TP + FP + TN + FN)	RQ1, RQ3
Precision / Recall	TP/(TP+FP) ; TP/(TP+FN)	RQ1, RQ3
F1	2(Precision × Recall)/(Precision + Recall)	RQ1, RQ3
ROC-AUC	Area under TPR-vs-FPR curve	RQ1, RQ3
McNemar's χ^2	$(\lvert b - c\rvert - 1)^2 / (b + c)$	RQ1, RQ3
RMSE	$\sqrt{[(1/n)\ \Sigma(\text{pred} - \text{actual})^2]}$	RQ2
QLIKE	$(1/n)\ \Sigma[(a/p) - \ln(a/p) - 1]$	RQ2
Sharpe ratio	(mean − rf)/sd, annualised × $\sqrt{252}$	RQ4

Preliminary Results

Preliminary Findings (by Research Question)

[Weeks 5–6 work. For each RQ, report what has been completed, what the preliminary outputs show (with references to Table/Figure numbers), and what remains. Begin with RQ1 (FinBERT vs. lexicon on the collected labelled corpus), which can be run first.]

Preliminary Model Performance

Table 11

Preliminary Model Performance Comparison

RQ	Model	Baseline metric	Augmented metric	Change
RQ1	FinBERT vs lexicon	[acc]	[acc]	[Δ]
RQ2	HAR + sentiment	[RMSE/QLIKE]	[RMSE/QLIKE]	[Δ]
RQ3	Technicals + sentiment	[acc/F1]	[acc/F1]	[Δ]
RQ4	Strategy vs buy-and-hold	[Sharpe]	[Sharpe]	[Δ]

Interim Limitations and Risks

- Free-tier data quota (Alpha Vantage, 25 requests/day) constrains the sentiment universe to eight tickers and slows collection; mitigated by caching and resumable, per-ticker pulls.
- StockTwits serves only recent messages, so it supports RQ1 validation rather than the historical panel; mitigated by using Alpha Vantage news as the historical sentiment feature.
- Sentiment effects are expected to be economically small (per the literature); mitigated by conservative effect-size assumptions in the sample-size design.
- Rural hotspot connectivity causes intermittent collection timeouts; mitigated by retry/backoff and the option to collect from a stable cloud environment.

Statistical power for RQ4 is limited: the strategy test requires 1,558 trading days and the study window provides 500, so the backtest is under-powered and a null result there is inconclusive rather than negative; mitigated by a block-bootstrap implementation of the Ledoit-Wolf test and by explicit reporting of the shortfall.

Effective sample size is materially smaller than the row count. The eight tickers move together, with a mean pairwise correlation of 0.359 in realized volatility, so the 3,840 ticker-days correspond to roughly 2.3 independent series and an effective sample nearer 1,093 observations; mitigated by reporting standard errors clustered by date and by sizing claims against the adjusted figure rather than the raw row count.

The study window contains a single volatility regime. Quarterly cross-sectional mean volatility varies only 1.25-fold across 2023-2024, with no crisis or sustained stress episode, so the models are trained and validated under calm to moderate conditions only; mitigated by restricting all claims to comparable conditions and identifying regime generalisation as future work.

Attention is measured from news-media coverage rather than retail activity. The direct retail measure would be StockTwits message volume, which is unavailable historically because the API serves only recent messages; mitigated by treating article count as a proxy and stating the substitution explicitly.

Next Steps (for Final Report)

- Complete Alpha Vantage news collection for all eight sentiment tickers.
- Assemble the final analysis panel and run data cleaning and EDA with interpretation.
- Train and validate FinBERT against the lexicon baseline (RQ1).
- Fit and evaluate the volatility (RQ2) and direction (RQ3) models out-of-sample.
- Run the cost-aware strategy backtest and significance test (RQ4).
- Report preliminary and final results, and draft the implementation, recommendation, and limitations sections.

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Appendix

[Optional. Include any code referenced in the text (e.g., feature-building or collection scripts), with each appendix titled and referenced from the body.]